

PROFILE-BASED SEQUENTIAL SELECTION

1 Assess graph

Profile s ; statistics g



Local scope

Isolated nodes
Redundant triples



Graph scope

Empty fields / loops
Invalid relations
Hierarchy-rule flags



Source scope

Unsupported values
Source-match rate



Trial evaluation criteria



Feasibility

Restoration bounds
Density / edit guards



Quality gain

Profile change
Hard-violation reduction



Utility terms

Edit cost, scope prior
and confidence

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Profile-conditioned routing

$$(p_{\text{repair}}, \pi) = f_{\phi}([s; g])$$

Proceed if $p_{\text{repair}} \geq \tau_{\text{repair}}$ OR a hard violation exists

Local prior / Graph prior / Source prior

No detected violations: stop

3

Construct candidate edits



Fixed model proposals

One edit bundle per relation



Rule proposals

Edits from detected violations



Delete



Retype



Add

Fixed costs: delete 0.30 > retype 0.16 > add 0.06

5

Commit one edit



Updated graph
+ decision log

4

Trial + select

Evaluate all
candidate trial graphs

Keep feasible edits
with positive utility

Select highest utility

None eligible:
stop; keep current graph

bounds + utility

Reassess the accepted graph; repeat within the iteration limit

Stop: ($p_{\text{repair}} < \tau_{\text{repair}}$ AND no hard violation), OR no detected violations,

OR no feasible positive-utility edit, OR iteration limit reached.